~~$V^{\pi}(s_1) = 0.6 \times 4 + 0.4 \times 1 + 1 \times 3 =$~~

1 deterministic policy

$$V^{\pi}(s_1) = \sum_a \pi(a_1|s_1) \sum_{s',r} p(s', r | s_1, a)\left[r + \gamma V^{\pi}(s')\right]$$

$$V^{\pi}(s_1) = 1 * \times \left[0.6 \times \left[5.8 + 0.9 \times V^{\pi}(s_1)\right] + 0.4 \times \left[0.6 * 0.9 \times V^{\pi}(s_2)\right]\right]$$

$$V^{\pi}(s_2) = 1 \times \left[0.2 \times \left[5.8 + 0.9 \times V^{\pi}(s_1)\right] + 0.8 \times \left[0.6 + 0.9 \times V^{\pi}(s_2)\right]\right]$$

$$\Rightarrow V^{\pi}(s_1) = 3.48 + 0.54 V^{\pi}(s_1) + 0.24 + 0.36 V^{\pi}(s_2)$$

$$\Rightarrow V^{\pi}(s_2) = 1.16 + 0.18 V^{\pi}(s_1) + 0.72 + 0.72 V^{\pi}(s_2)$$

$$\Rightarrow 0.46 x - 0.36 y = 3.72$$

$$\Rightarrow -0.18 x + 0.28 y = 1.64$$

$$\Rightarrow x = 25.5 \qquad y = 22.25$$

$$\Rightarrow V^{\pi}(s_1) = 25.5 \qquad V^{\pi}(s_2) = 22.25$$